AeroSaga Run Commands

Use **three PowerShell windows**.

PowerShell 1: Start Docker and Temporal

Start Docker Desktop:

Start-Process "$env:LOCALAPPDATA\Programs\DockerDesktop\Docker Desktop.exe"

Check Docker:

docker info

You must see a Server: section.

Start Temporal:

Set-Location "D:\AeroSaga"

docker compose up -d

docker compose ps

Expected status:

Up ... (healthy)

If the docker command is not recognized, use the full Docker path:

$docker = "$env:LOCALAPPDATA\Programs\DockerDesktop\resources\bin\docker.exe"

Set-Location "D:\AeroSaga"

& $docker compose up -d

& $docker compose ps

Open the Temporal dashboard:

http://localhost:8233

PowerShell 2: Start the backend

Open a new PowerShell window:

Set-Location "D:\AeroSaga\backend"

mvn spring-boot:run

Wait for:

Tomcat started on port 8080

Started AeroSagaApplication

Keep this window open.

The backend uses:

http://localhost:8080

PowerShell 3: Start the frontend

Open another new PowerShell window:

Set-Location "D:\AeroSaga\frontend"

npm install

npm run dev

npm install is normally needed only the first time.

Open the website:

http://127.0.0.1:5173/

or:

http://localhost:5173/

Keep this window open.

Test the backend directly

Open a fourth PowerShell window:

```
$body = @{
   droneId = "DRONE-001"
} | ConvertTo-Json

Invoke-RestMethod `
   -Method Post `
   -Uri "http://localhost:8080/api/missions/takeoff" `
   -ContentType "application/json" `
   -Body $body
```

Expected result:

```
status  : COMPLETED
message : Drone DRONE-001 takeoff simulated successfully
```

Test a workflow status

Save a workflow ID from the previous response:

```
$workflowId = "drone-takeoff-drone-001-YOUR-UUID"
```

Then run:

```
Invoke-RestMethod `
   -Method Get `
   -Uri http://localhost:8080/api/missions/takeoff/$workflowId
```

Use the frontend

Open:

http://127.0.0.1:5173/

Then:

1. Select a drone.
2. Click **SIMULATE TAKEOFF**.
3. Wait for the result.

Expected:

COMPLETED

Drone DRONE-001 takeoff simulated successfully

Check service status

Docker and Temporal

Set-Location "D:\AeroSaga"

docker compose ps

Backend port

Test-NetConnection localhost -Port 8080

Expected:

TcpTestSucceeded : True

Frontend port

Test-NetConnection localhost -Port 5173

Expected:

TcpTestSucceeded : True

Temporal dashboard

http://localhost:8233/

Development validation commands

Backend tests

Set-Location "D:\AeroSaga\backend"

mvn test

Frontend type check

Set-Location "D:\AeroSaga\frontend"

npm run lint

Frontend production build

Stop the project

Stop the backend and frontend by pressing:

Ctrl+C

in their PowerShell windows.

Then stop Temporal:

Set-Location "D:\AeroSaga"

docker compose down

Docker Desktop can remain open or be closed separately.

Complete daily startup sequence

Start-Process "$env:LOCALAPPDATA\Programs\DockerDesktop\Docker Desktop.exe"

```
docker info

Set-Location "D:\AeroSaga"
docker compose up -d
docker compose ps

Start-Process "$env:LOCALAPPDATA\Programs\DockerDesktop\Docker Desktop.exe"

docker info

Set-Location "D:\AeroSaga"
docker compose up -d
docker compose ps

Set-Location "D:\AeroSaga\backend"
mvn spring-boot:run

Set-Location "D:\AeroSaga\backend"
mvn spring-boot:run

In a third PowerShell window:
Set-Location "D:\AeroSaga\frontend"
npm run dev

Then open:
http://127.0.0.1:5173/
```